

TECHNICAL DATA SHEET (TDS) – PRODUCT 1 (THE REALITY)

PRODUCT NAME: Heavy-Duty Aluminum-Core Vulcanized Intake Manifold Gasket Set

APPLICATION: BMW E38 750i / E31 850Ci – M73B54 V12 Engine (High-Heat V-Bank Application)

COMPONENTS: 2x Inner/Center Gaskets + 2x Outer Gaskets (4-piece full set per engine)

OEM CROSS-REFERENCES:

- **Inner / Center Position:** 11611736656
- **Outer Position:** 11611736657

1. STRUCTURE & GEOMETRY

- **Manufacturing Process: Rubber-to-Metal Injection Over-Molding.** The elastomer is permanently bonded and vulcanized over a structural light-alloy core.
- **Core Design:** Single-piece integrated aluminum frame with machined structural guide studs and compression-limiting shoulder sleeves.
- **Asymmetry:** Left-to-Right and Inner-to-Outer asymmetrical 3-port configuration according to exact OEM M73 spatial constraints.

2. MATERIAL SPECIFICATIONS

- **Internal Structural Core: Aluminium Alloy (AlSi9Cu3 or AA380 / AA5754)** for structural rigidity, zero thermal warpage, and thermal expansion matching with the M73 cylinder head.
- **Vulcanized Outer Coating & Sealing Lips: Premium High-Fluorine FKM (Viton® Type B or Type F)** according to **ASTM D1418 / ISO 1629**.
- **Thermal Window:** Continuous operational resistance: **-40°C to +230°C** (Short-term peak threshold: **+275°C**). Optimized for the extreme trapped heat inside the V12 engine valley.
- **Chemical Stability:** 100% immune to degradation from premium unleaded fuels, high-concentration ethanol (E15/E85), engine blow-by acids, and synthetic engine oils.
- **Bonding Agent (Metal-to-Rubber):** High-heat vulcanization primer (**Chemlok® / Megum™** system) baked before injection molding.

3. CRITICAL PERFORMANCE METRICS

- **Delamination Resistance:** Must pass **ISO 2409 (Cross-cut test)** Class 0 and **ASTM D429 (Method B)** rubber-to-metal peel strength test. Zero separation after 500 hours of continuous hot oil (150°C) and fuel vapor testing.
- **Plane Flatness Tolerance:** < 0.03 mm structural flatness across the entire block surface to ensure a absolute zero-leak seal, protecting the V12 bank-to-bank electronic synchronization.



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